Theory & Manual

for

AlkCalc

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I. Introduction

Bound single-electron eigenfunctions of alkali-metal atoms and alkaline-earth metal ions are of significant relevance in quantum computing, quantum information processing, quantum simulation, Rydberg physics, quantum optics, ... However, except for the special case of the Hydrogen atom and hydrogenic ions, it is not possible to solve the time-independent Schrödinger equation analytically. Because of this, there is the need to obtain eigenenergies and eigenstates numerically. The library `AlkCalc` is designed for exactly this purpose: it allows the user to compute the relevant part of the bound-state energy spectrum as well as the associated eigenstates.

The software `AlkCalc` itself is split into two parts: the first part is used to compute the eigenenergies and eigenstates *once* and store the resulting data. The second part are the *library functions*. These functions implement useful functionality for reading and processing the precomputed atomic structure data. For instance, there are functions to compute radial transition matrix elements and oscillator strengths.

In the various README files included in the file tree of `AlkCalc` , technical details for computing the eigenenergies and eigenstates are provided. However, these files do not include a discussion on the theory and physics behind the code. Further, there is no reference manual for the library functions. This document's purpose is to add the missing pieces: in Sec. II the theory, the physics, and the numerical method associated with `AlkCalc` is discussed. Section III is a comprehensive reference manual for the library functions of `AlkCalc` .

II. Theory

This section is organised as follows: in Subsec. i the theory and the physics of single-atom/ion systems is discussed. In Subsec. ii the numerical method based on B-splines is presented. In Subsec. iii it is shown how radial transition matrix elements are computed by `AlkCalc` . In Subsec. iv it is shown how oscillator strengths for electronic transitions are computed. Finally, in Subsec. v it is shown how lifetimes of electronic states are computed by `AlkCalc` .

i. Single-atom/ion theory

The starting point is an alkali-metal atom or alkaline-earth-metal ion. In the ground-state of these systems all electrons are in closed shells except for one valence electron. Examples of such systems are ^{1}H, ^{85}Rb, ^{88}Sr^{+}. This property allows a Kepler-problem description for the 'effective nucleus' (the nucleus 'screened' by the closed-shell electrons) and the valence electron. The dynamics of the valence electron and the effective nucleus can be described with the following Hamiltonian:

$$H = \frac{\boldsymbol{p}^2}{2m} + V(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2)\,. \tag{II.1}$$

Here, $m = Mm_\mathrm{e}/(M + m_\mathrm{e})$ is the reduced mass of the system, composed of the electron's mass m_e and the atom's or ion's mass M. The centre-of-mass motion is decoupled and $\boldsymbol{r}, \boldsymbol{p}$ are the relative canonical co-ordinates, i.e., $\boldsymbol{r}$ is the distance vector between the centre-of-mass position of the effective nucleus and the position of the valence electron. Further, $\boldsymbol{L}$, $\boldsymbol{S}$, and $\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}$

are the orbital angular momentum operator, the electron's spin operator, and the total angular momentum operator, respectively. Finally, the radially symmetric potential V, depending on the distance $r = \|\boldsymbol{r}\|$, describes the interaction between the valence electron and the effective nucleus. It accounts for several effects: far from the nucleus the valence electron experiences a Coulomb potential with effective charge $Z_\mathrm{c} = 1$ for alkali-metal atoms and $Z_\mathrm{c} = 2$ for singly ionised alkaline-earth metal atoms. This is because the nucleus of charge Z is 'screened' by the $Z - 1$ (or $Z - 2$, for alkaline-earth metal ions) ground-state electrons in closed-shells. However, valence electrons with orbits sufficiently close to the nucleus experience the non-trivial charge distribution posed by the nucleus in conjugation with the tightly bound electron cloud. This leads to short-ranged corrections to the effective Coulomb potential with charge Z_c at long distances. Following Refs. [1–5], these corrections are described by the following *parametric model potential*:

$$V(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2) = V_\mathrm{C}(r) + V_\mathrm{P}(r) + V_\mathrm{R}(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2)\,. \tag{II.2}$$

The first contribution is

$$V_\mathrm{C}(r) = -\frac{e^2}{4\pi\varepsilon_0}\frac{Z_\mathrm{n}(r)}{r}\,,\; Z_\mathrm{n}(r) = Z_\mathrm{c} + (Z - Z_\mathrm{c})\mathrm{e}^{-k_1 r} + k_2 r\mathrm{e}^{-k_3 r} - k_4 r^2\mathrm{e}^{-k_3 r}\,. \tag{II.3}$$

This is a modified Coulomb potential which has the tail of a Coulomb potential with charge Z_c and short-ranged corrections for small radii accounting for the charge-distribution of the effective nucleus. Here, the parameters k_j ($j = 1, 2, 3, 4$) are empirically determined from spectroscopic data [1, 2]. The second contribution to the potential (II.2) is

$$V_\mathrm{P}(r) = -\frac{e^2}{(4\pi\varepsilon_0)^2}\frac{\alpha_\mathrm{d}}{2r^4}\left[1 - \exp\left(-\frac{r^6}{r_\mathrm{c}^6}\right)\right]. \tag{II.4}$$

It accounts for the polarisability of the screened nucleus, which gets polarised by the presence of the valence electron. Here, r_c is the effective nucleus' size and α_d its static electric polarisability. Finally,

$$V_\mathrm{R}(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2) = \frac{1}{2m^2c^2}\frac{V'_\mathrm{NR}(r)}{rN(r)}\boldsymbol{L}\cdot\boldsymbol{S}\,,\; N(r) = \left[1 - \frac{V_\mathrm{NR}(r)}{2mc^2}\right]^2\,,\; V_\mathrm{NR}(r) = V_\mathrm{C}(r) + V_\mathrm{P}(r)\,. \tag{II.5}$$

It describes (relativistic) coupling between the orbital angular momentum and the valence electrons' spin. This expression results from relativistic perturbation theory [3]. Here, V_NR is the non-relativistic model potential and c the speed of light. For expressing the spin-operator $\boldsymbol{S}$ in Eq. (II.5) the representation in Pauli's (spin) matrices is chosen:

$$\boldsymbol{S} = \frac{\hbar}{2}\begin{bmatrix}\sigma_x\\ \sigma_y\\ \sigma_z\end{bmatrix}. \tag{II.6}$$

The Pauli spin matrices are

$$\sigma_x = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix},\; \sigma_y = \begin{bmatrix}0 & -\mathrm{i}\\ \mathrm{i} & 0\end{bmatrix},\; \sigma_z = \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}. \tag{II.7}$$

This simple choice for representing the spin degrees of freedom is possible because throughout this work the interest is in the case where the spin is $s = 1/2$.

At this point the Hamiltonian which determines the eigenenergies and the eigenstates is defined. In the following, the goal is to cast the eigenvalue problem posed by Hamiltonian (II.1) into a more practical form for the numerical procedure. The first step is to rewrite the product $\boldsymbol{L} \cdot \boldsymbol{S}$ in terms of the total angular momentum operator $\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}$:

$$\boldsymbol{L} \cdot \boldsymbol{S} = \frac{1}{2}\left[\boldsymbol{J}^2 - \boldsymbol{L}^2 - \boldsymbol{S}^2\right]. \tag{II.8}$$

Next, the square of the momentum operator in spherical co-ordinates is rewritten:

$$\boldsymbol{p}^2 = (-\mathrm{i}\hbar\nabla)^2 = -\hbar^2\nabla^2 = -\frac{\hbar^2}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{\boldsymbol{L}^2}{r^2}. \tag{II.9}$$

The second contribution is the centrifugal barrier. In a classical Kepler problem this is what prevents the electron crashing into the nucleus. Expressed in the spherical coordinates, Hamiltonian (II.1) takes the form

$$H = -\frac{\hbar^2}{2mr^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + V_{\mathrm{eff}}(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2), \;\; V_{\mathrm{eff}}(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2) = V(r; \boldsymbol{L}^2, \boldsymbol{S}^2, \boldsymbol{J}^2) + \frac{\boldsymbol{L}^2}{2mr^2}. \tag{II.10}$$

The full Hilbert space this Hamiltonian acts on can be split into an angular momentum factor and a radial one. A basis of the angular momentum Hilbert space is given by the simultaneous eigenstates Φ_{l,s,j,m_j} of the operators (Cartan subalgebra) $\boldsymbol{L}^2$, $\boldsymbol{S}^2$, $\boldsymbol{J}^2$, and J_z. Note that a simultaneous eigenbasis of $\boldsymbol{L}^2$, $\boldsymbol{S}^2$, $\boldsymbol{J}^2$, and J_z exists, because these operators commute pairwise. The so-called *spinors* $\boldsymbol{\Phi}_{l,s,j,m_j}$ constitute the so-called *coupled basis*, also known as the *fine-structure* basis in the literature. These spinors are given by linear combinations of the uncoupled basis states

$$Y_{l,m_l} \otimes \boldsymbol{\chi}_{s,m_s} \tag{II.11}$$

via SU(2) Clebsch-Gordan coefficients [6]. Here, $Y_{l,m_l} : [0,\pi]\times[0,2\pi] \to \mathbb{C}$ are spherical harmonics defined as

$$Y_{l,m_l}(\vartheta,\varphi) = \sqrt{\frac{2l+1}{4\pi}\frac{(l-m_l)!}{(l+m_l)!}}P_l^{m_l}(\cos(\vartheta))\,\mathrm{e}^{\mathrm{i}m_l\varphi}. \tag{II.12}$$

In this expression $P_l^{m_l} : [-1,1] \to \mathbb{R}$ are the associated Legendre polynomials defined for $l = 0, 1, 2, \ldots$ and $m_l = 0, 1, \ldots, l-1, l$. These polynomials can be computed via the recursion

$$(l-m_l)P_l^{m_l}(x) = x(2l-1)P_{l-1}^{m_l}(x) - (l+m_l-1)P_{l-2}^{m_l}(x), \tag{II.13}$$

with starting point

$$P_{m_l}^{m_l}(x) = (-1)^{m_l}\frac{(2m_l)!}{2^{m_l}m_l!}(1-x^2)^{m_l/2}, \tag{II.14}$$

in combination with $P_l^{m_l}(x) = 0$ for $l < m_l$. Negative values of m_l are treated with the relation

$$P_l^{-m_l} = (-1)^{m_l}\frac{(l-m_l)!}{(l+m_l)!}P_l^{m_l}. \tag{II.15}$$

In this definition of the associated Legendre polynomial the so-called *Condon-Shortley phase* is included. This follows the definition in Ref. [6].[1] Finally, $\boldsymbol{\chi}_{s,m_s}$ in Eq. (II.11) are two-spinors defined as

$$\boldsymbol{\chi}_{\frac{1}{2},\frac{1}{2}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \ \boldsymbol{\chi}_{\frac{1}{2},-\frac{1}{2}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{II.16}$$

Explicitly (for the case of interest $s = 1/2$) the coupled basis states $\boldsymbol{\Phi}_{l,s,j,m_j}$ are:

$$\boldsymbol{\Phi}_{l,s,j,m_j} = \langle Y_{l,m_j-s} \otimes \boldsymbol{\chi}_{s,s}, \boldsymbol{\Phi}_{l,s,j,m_j} \rangle Y_{l,m_j-s} \otimes \boldsymbol{\chi}_{s,s} + \langle Y_{l,m_j+s} \otimes \boldsymbol{\chi}_{s,-s}, \boldsymbol{\Phi}_{l,s,j,m_j} \rangle Y_{l,m_j+s} \otimes \boldsymbol{\chi}_{s,-s}\,, \tag{II.17}$$

where the scalar products are (possibly vanishing) SU(2) Clebsch-Gordan coefficients. These spinors transform according to irreducible representations of SU(2). This is seen in the following way: consider the unitary and faithful representation $\rho(\boldsymbol{\alpha}) = \exp(\mathrm{i}\boldsymbol{\alpha} \cdot \boldsymbol{J})$, where $\boldsymbol{\alpha} \in \mathbb{R}^3$ is a vector of length $\|\boldsymbol{\alpha}\| \in [0, 2\pi)$. Its action on the spinors is

$$\rho(\boldsymbol{\alpha})\Phi_{l,s,j,m_j}(\vartheta, \varphi) = \sum_{q=1}^{d_j} D^{(\Gamma_j)}(\boldsymbol{\alpha})_{m_j,q}\Phi_{l,s,j,q}(\vartheta, \varphi)\,. \tag{II.18}$$

In this expression, d_j is the dimension of the irreducible representation Γ_j of SU(2) and $D^{(\Gamma_j)}$ is a unitary matrix representation of the irreducible representation Γ_j.

To rewrite the Hamiltonian the crucial step is to use the fact that it has SU(2) symmetry, i.e., $[\rho(\boldsymbol{\alpha}), H] = 0$ for all $\boldsymbol{\alpha}$.[2] Due to this symmetry the Hamiltonian is automatically block diagonal with block dimensions given by integer multiplicities μ_j of the dimensions $d_j = 2j + 1$ of the irreducible representations Γ_j labelled by $j = 0, 1/2, 1, 3/2, 2, \ldots$. The multiplicities μ_j are $\mu_j = 2$ for $j \neq 0$ and half-integer and $\mu_j = 0$ for all integer j including $j = 0$. This is because for each $l > 0$ there are the two possible orbital angular momentum quantum numbers $l = j - 1/2$ and $l = j + 1/2$ and for $s = 1/2$ a total angular momentum of $j = 0$ is not possible. This entails that the Hamiltonian in the basis of the spinors $\boldsymbol{\Phi}_{l,s,j,m_j}$ can be decomposed into the following direct sum:

$$\bigoplus_{l=0}^{\infty} \bigoplus_{j\in\{|l-s|,l+s\}} \bigoplus_{m_j=-j}^{j} H_{l,s,j,m_j}\,, \quad H_{l,s,j,m_j} = \langle \boldsymbol{\Phi}_{l,s,j,m_j}, H\boldsymbol{\Phi}_{l,s,j,m_j} \rangle\,. \tag{II.19}$$

Note that the off-diagonal matrix elements within the blocks associated with each irreducible representation Γ_j vanish, i.e., $\langle \boldsymbol{\Phi}_{l,s,j,m_j'}, H\boldsymbol{\Phi}_{l,s,j,m_j} \rangle = 0$ for $m_j' \neq m_j$. This is a result of the fact that although the multiplicities μ_j can be larger than 1, the additional conservation of ℓ enforces that the each j-block is still proportional to the identity on the angular Hilbert space, as must be the case according to Schur's Lemma. Therefore, the full Hamiltonian (II.1) in the basis of the spinors $\boldsymbol{\Phi}_{l,s,j}$ can be expressed as the more simple direct sum

$$\bigoplus_{l=0}^{\infty} \bigoplus_{j\in\{|l-s|,l+s\}} 1_{2j+1} \otimes H_{l,s,j}\,. \tag{II.20}$$

1 This definition entails that the factor $(-1)^{m_l}$ does not explicitly appear in the definition of the spherical harmonics. This remark is placed here to avoid confusion of whether the spherical harmonics are defined with or without the Condon-Shortley phase. In the definition of Y_{l,m_l} used in this work, the Condon-Shortley phase included.

2 The symmetry group is even bigger because parity is also conserved. So the full symmetry group is $\mathrm{SU}(2) \times \mathbb{Z}_2$.

Here, 1_{2j+1} is the identity matrix of dimension $2j+1$ and

$$H_{l,s,j} = -\frac{\hbar^2}{2mr^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + V_{\text{eff}}(r;\hbar^2 l[l+1],\hbar^2 s[s+1],\hbar^2 j[j+1])\,. \tag{II.21}$$

For completeness, ρ can be expressed in the basis of $\mathbf{\Phi}_{l,s,j}$ as well. It is given by the following direct sum of the irreducible representations Γ_j is SU(2):

$$\rho = \bigoplus_{\{j\,:\,2j\in\mathbb{N}_{>0}\}} 2\Gamma_j\,. \tag{II.22}$$

The block structure of the Hamiltonian allows to reduce the eigenvalue problem posed by the time-independent Schrödinger equation

$$H\mathbf{\Psi} = E\mathbf{\Psi} \tag{II.23}$$

to the blocks $H_{l,s,j}$ on the Hilbert space associated with the radial degrees of freedom. In particular, solving for the eigenenergies and the eigenstates is reduced to solving the radial eigenvalue problems

$$H_{l,s,j}R_{n,l,s,j} = E_{n,l,s,j}R_{n,l,s,j}\,, \tag{II.24}$$

where $R_{n,l,s,j}$ is the n-th radial eigenstate of $H_{l,s,j}$. The total eigenstate $\mathbf{\Psi}_{n,l,s,j,m_j}$ is then given by the product of the radial eigenstates and the spinors:

$$\mathbf{\Psi}_{n,l,s,j,m_j}(r,\vartheta,\varphi) = R_{n,l,s,j}(r)\mathbf{\Phi}_{l,s,j,m_j}(\vartheta,\varphi)\,. \tag{II.25}$$

This shows that solving for the full eigenstates $\mathbf{\Psi}_{n,l,s,j,m_j}$ is equivalent to solving for the radial eigenstates $R_{n,l,s,j}$. In general, the eigenenergies $E = E_{n,l,s,j}$ and the radial eigenstates $R_{n,l,s,j}$ must be calculated numerically. This task is exactly what `AlkCalc` is designed to do.

ii. Numerical method

In this subsection the numerical method which is employed by `AlkCalc` to solve the radial eigenvalue problems posed by Eq. (II.24) for the eigenenergies and the radial eigenstates $R_{n,l,s,j}$ is formulated. The first step is to express the eigenvalue problems in terms of dimensionless quantities. To this end the Hamiltonians $H_{l,s,j}$ in Eq. (II.21) are expressed in units of Hartree [6], $E_{\text{H}} = e^2/(4\pi\varepsilon_0 a_{\text{B}})$, where $a_{\text{B}} = \hbar/(m_{\text{e}}c\alpha) \approx 52.9\,\text{pm}$ is Bohr's radius. The radial eigenstates are parametrised as

$$R_{n,l,s,j}(r) = \frac{1}{\sqrt{a_{\text{B}}^3}}\frac{f_{n,l,s,j}(r/a_{\text{B}})}{r/a_{\text{B}}}\,, \tag{II.26}$$

where $f_{n,l,s,j}$ is a dimensionless eigenfunction. The eigenvalue problems posed by Eq. (II.24) expressed in the dimensionless eigenfunctions (II.26) take the form

$$\left[-\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}t^2} + \frac{l(l+1)}{2t^2} + C\frac{V(a_{\text{B}}t;\hbar^2 l[l+1],\hbar^2 s[s+1],\hbar^2 j[j+1])}{E_{\text{H}}}\right] f_{n,l,s,j}(t) = \bar{E}_{n,l,s,j}f_{n,l,s,j}(t)\,, \tag{II.27}$$

where

$$C = \frac{1}{1 + m_\mathrm{e}/M} \tag{II.28}$$

and the eigenenergies may be extracted from the relation

$$\frac{E_{l,s,j,n}}{E_\mathrm{H}} = \frac{\bar{E}_{n,l,s,j}}{C}\,. \tag{II.29}$$

Rewriting the normalisation condition in terms of the dimensionless eigenfunctions $f_{n,l,s,j}$ yields

$$1 = \int_0^\infty \mathrm{d}r\, r^2 R^2_{n,l,s,j}(r) = \int_0^\infty (a_\mathrm{B}\,\mathrm{d}t)(a_\mathrm{B}t)^2 \frac{1}{a_\mathrm{B}^3}\frac{f^2_{n,l,s,j}(t)}{t^2} = \int_0^\infty \mathrm{d}t\, f^2_{n,l,s,j}(t)\,. \tag{II.30}$$

Here it is assumed that the radial eigenfunctions are real-valued, which is justified as the Hamiltonian is not only an hermitian but also a symmetric operator. In conclusion, normalising the dimensionless eigenfunctions $f_{n,l,s,j}$ with respect to the standard one-dimensional L^2-norm automatically yields the correctly normalised radial eigenstates.

The ultimate goal is to express Eq. (II.27) as a matrix eigenvalue problem which can be directly addressed with a computer. To obtain a matrix-eigenvalue problem the first step is to choose a basis in which the dimensionless eigenfunctions can be represented. There are many different sets of basis functions to choose from. In the case of `AlkCalc` so-called *B-splines* [7] are chosen.[3] B-splines are piecewise polynomial functions with finite support. In the following it is first discussed how these B-splines, i.e., the basis functions, are constructed. Subsequently, the matrix eigenvalue problems are formulated.

The first step in constructing the B-splines is to truncate the domain of the radial eigenfunctions $f_{n,l,s,j}$ from $\mathbb{R}_{\geq 0}$ to a finite interval $I = [0, t_\mathrm{max}] \subset \mathbb{R}_{\geq 0}$, where $t_\mathrm{max} > 0$ defines the maximally considered radius $r_\mathrm{max} = a_\mathrm{B} t_\mathrm{max}$. This truncation is justified whenever the maximal t_max is chosen sufficiently large such that the support of the radial eigenfunctions $f_{n,l,s,j}$ of interest lays within the interval I. Strictly speaking, the support of the radial eigenstates is not finite, as they typically approach zero only asymptotically. However, in practice the support can be considered finite whenever the value of the radial eigenstates beyond a certain value of r (or equivalently t) stays below machine precision. The next step in the construction of the B-splines is to choose a family of points $(t_i)_{i=0,\ldots,N_\mathrm{ks}-1}$, the so-called *knots*, in the interval I, where $N_\mathrm{ks} = N - 1 + 2d$ is the total number of *knots*.[4] Here, $N \in \mathbb{N}$ is the number of unique knots and $d \geq 0$ is the degree of the B-splines: each B-spline is a piecewise polynomial function; a degree-d B-spline is a B-spline where all these polynomials have (maximally) degree d. The knots t_i are defined as

$$t_i = \begin{cases} 0 & , \quad 0 \leq i \leq k-1 \\ t_i & , \quad k \leq i \leq k+N-3 \\ t_\mathrm{max} & , \quad k+N-2 \leq i \leq 2k+N-3 \end{cases}, \tag{II.31}$$

3 This method may be conceptually and programmatically somewhat more involved compared to a standard finite-difference or finite-element method, however, it is preferred here for two reasons: first, B-splines allow to work with much smaller matrix dimensions and therefore allow to efficiently solve the associated matrix-eigenvalue problems on a computer. Second, the small matrix dimensions allow to space-efficiently store the computed radial eigenstates.

4 The family $(t_i)_{i=0,\ldots,N_\mathrm{ks}-1}$ is sometimes called the *knot vector* in the literature.

where the so-called $N-2$ *internal knots* $t_k < \cdots < t_{k+N-3}$ are chosen in strictly ascending order and $t_0 < t_k$ and $t_{k+N-3} < t_{\max}$. This construction can be visualised as follows:

$$\underbrace{t_0, \dots, t_0}_{k \text{ times}}, \underbrace{t_k, \dots, t_{k+N-3}}_{\text{Internal knots}}, \underbrace{t_{\max}, \dots, t_{\max}}_{k \text{ times}} . \tag{II.32}$$

There are N different knots. The first and the last one are repeated $k = d+1$ times, where k is called the *order* of the B-splines. On the knots $(t_i)_{i=0,\dots,N_{\text{ks}}-1}$ the $N_{\text{bs}} = N + d - 1$ many B-splines $u_i^{(d)}$ of degree d (or order k) are defined via de Boor's recursion formula [7] ($q = 1, \dots, d$, $i = 0, \dots, N_{\text{bs}} - 1$):

$$u_i^{(q)}(t) = \frac{t - t_i}{t_{i+q} - t_i} u_i^{(q-1)}(t) + \frac{t_{i+q+1} - t}{t_{i+q+1} - t_{i+1}} u_{i+1}^{(q-1)}(t), \quad u_i^{(0)}(t) = \begin{cases} 1 & , \quad t_i \le t < t_{i+1} \\ 0 & , \quad \text{else} \end{cases} . \tag{II.33}$$

The support of the resulting B-splines is restricted to the interval $[t_i, t_{i+k})$. One can show that with the chosen initial condition in Eq. (II.33) all B-splines $u_i^{(d)} : \mathbb{R} \to [0,1]$, i.e., map into the interval $[0,1]$.[5] Further one shows that

$$\sum_{i=0}^{N_{\text{bs}}-1} u_i^{(d)} = 1 . \tag{II.34}$$

These properties entail that the basis of B-splines

$$\left(u_0^{(d)}, \dots, u_{N_{\text{bs}}-1}^{(d)}\right) \tag{II.35}$$

is a partition of unity. With this normalisation the B-spline basis is uniquely defined for the choice of knots $(t_i)_{i=0,\dots,N_{\text{ks}}-1}$. To numerically evaluate the B-splines `A1kCa1c` employs the double precision version of de Boor's algorithm which is implemented by the *FORTRAN* routine DBSPVD [8] taken from *Netlib* [9].

With the B-spline basis at hand, the next step is to express the eigenvalue problem posed by Eq. (II.27) in that basis. To this end one integrates over the eigenvalue problem with a test function g to reduce it to its so-called *weak form*:

$$\int_{-\infty}^{\infty} \mathrm{d}t\, g(t) \left(-\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}t^2} + v(t) \right) f(t) = \lambda \int_{-\infty}^{\infty} \mathrm{d}t\, g(t) f(t) . \tag{II.36}$$

To simplify the notation the indices on $f_{n,l,s,j}$ are suppressed and the dimensionless eigenenergy renamed to $\lambda = \bar{E}_{n,l,s,j}$. Further, the potential terms in Eq. (II.27) are collected into the function

$$v(t) = \frac{l(l+1)}{2t^2} + C \frac{V(a_{\text{B}} t; \hbar^2 l[l+1], \hbar^2 s[s+1], \hbar^2 j[j+1])}{E_{\text{H}}} . \tag{II.37}$$

To reduce the weak form of the eigenvalue problem to a matrix eigenvalue problem one expands the eigenfunctions f in the B-spline basis:

$$f \approx \sum_{i=0}^{N_{\text{bs}}-1} f_i u_i^{(d)} . \tag{II.38}$$

5 The B-splines are defined on $\mathbb{R}$ with the understanding that each B-spline is zero outside the interval I.

In the following, $\boldsymbol{f} = (f_1, \dots, f_{N_{\mathrm{bs}}-1})^{\mathrm{T}}$ is the vector containing the expansion coefficients f_i. It is important to note that this is an approximation, because in general one must employ an infinite basis to exactly represent the eigenfunctions f. A brief discussion regarding this approximation is reserved for later in this section. For now, the expansion of the eigenfunctions f given in Eq. (II.38) can be plugged into Eq. (II.36). With the choice $g = u_i^{(d)}$ one arrives at the expression

$$\sum_{j=0}^{N_{\mathrm{bs}}-1} f_j B\left(u_i^{(d)}(t), -\frac{1}{2}(u_j^{(d)})''(t) + v(t)u_j^{(d)}(t)\right) = \lambda \sum_{i=0}^{N_{\mathrm{bs}}-1} f_j B(u_i^{(d)}(t), u_j^{(d)}(t))\,. \tag{II.39}$$

Here, for convenience, the bilinear form

$$B(\alpha, \beta) = \int_I \mathrm{d}t\; \alpha(t)\beta(t) \tag{II.40}$$

was introduced for any two sufficiently well-behaved functions $\alpha, \beta : I \to \mathbb{R}$. Next, the homogeneous Dirichlet boundary conditions $f(0) = f(t_{\max}) = 0$ are enforced on f in Eq. (II.38). To this end one notices that for the choice of knots employed in this work the de Boor algorithm [7] is defined such that $u_i^{(d)}(0) = \delta_{i,0}$ and $u_i^{(d)}(0) = \delta_{i,N_{\mathrm{bs}}-1}$. This entails that $f(0) = f_0$ and $f(t_{\max}) = f_{N_{\mathrm{bs}}-1}$. As a result, the boundary conditions are uniquely enforced by $f_0 = f_{N_{\mathrm{bs}}-1} = 0$. Employing this result allows to eliminate the boundary terms in the sums in Eq. (II.39):

$$\sum_{j=1}^{N_{\mathrm{bs}}-2} f_j B\left(u_i^{(d)}(t), -\frac{1}{2}(u_j^{(d)})''(t) + v(t)u_j^{(d)}(t)\right) = \lambda \sum_{i=1}^{N_{\mathrm{bs}}-2} f_j B(u_i^{(d)}(t), u_j^{(d)}(t))\,. \tag{II.41}$$

To finally bring the eigenvalue problem posed by Eq. (II.27) into matrix form, the following matrices shall be defined: first, the so-called *stiffness* matrix K, which is a $(N_{\mathrm{bs}}-2)\times(N_{\mathrm{bs}}-2)$-dimensional matrix with components $(i, j = 1, \dots, N_{\mathrm{bs}} - 2)$ defined as

$$K_{i,j} = -\frac{1}{2}B((u_i^{(d)})'', (u_j^{(d)})) = \frac{1}{2}B((u_i^{(d)})', (u_j^{(d)})')\,. \tag{II.42}$$

The second equal sign is obtained by partial integration with the fact that for all indices i, j the B-splines satisfy $u_{i(j)}^{(d)}(0) = u_{i(j)}^{(d)}(t_{\max}) = 0$. The next matrix is the so-called *mass* matrix M.[6] This $(N_{\mathrm{bs}}-2)\times(N_{\mathrm{bs}}-2)$-dimensional matrix accounts for the fact that the B-splines are not orthogonal with respect to the bilinear form B in Eq. (II.40). Its components $(i, j = 1, \dots, N_{\mathrm{bs}} - 2)$ are defined as

$$M_{i,j} = B(u_i^{(d)}, u_j^{(d)})\,. \tag{II.43}$$

The last matrix W represents the potential. The components of this $(N_{\mathrm{bs}}-2)\times(N_{\mathrm{bs}}-2)$-dimensional matrix $(i, j = 1, \dots, N_{\mathrm{bs}} - 2)$ are defined as

$$W_{i,j} = B(u_i^{(d)}, v u_j^{(d)})\,. \tag{II.44}$$

Importantly, all three matrices K, M, and W are real, symmetric, and banded. The banded structure is a consequence of the finite support of the B-splines. Combined, these properties allow

6 The terminology 'stiffness' and 'mass' matrix is common terminology in literature on the finite-element method.

to employ efficient specialised numerical methods to solve for the coefficient vectors $\boldsymbol{f}$ and the eigenvalues λ.

Expressed in the three matrices K, M, and W the eigenvalue problem posed by Eq. (II.41) takes the final form

$$H\bar{\boldsymbol{f}} = \lambda M\bar{\boldsymbol{f}}\,, \quad H = K + W\,, \tag{II.45}$$

where $\bar{\boldsymbol{f}} = (f_1, \ldots, f_{N_{\mathrm{bs}}-2})^{\mathrm{T}}$ is the $(N_{\mathrm{bs}} - 2)$-dimensional coefficient vector which defines the full coefficient vector $\boldsymbol{f} = (0, \bar{\boldsymbol{f}}^{\mathrm{T}}, 0)^{\mathrm{T}}$ including the coefficients $f_1 = f_{N_{\mathrm{bs}}-1} = 0$. Therefore, a solution to the generalised matrix eigenvalue problem problem posed by Eq. (II.45) is equivalent to obtaining the full coefficient vector $\boldsymbol{f}$ and thereby an approximate solution to the eigenfunctions of the exact eigenvalue problem posed by Eq. (II.27). Note that the conversion formula in Eq. (II.26) finally relates this result directly to the radial eigenstates $R_{n,l,s,j}$. The normalisation of the eigenfunctions f is trivially achieved by normalising the coefficient vector with respect to the scalar product B represented by the positive definite mass matrix M, i.e., in order to normalise f one must normalise $\bar{\boldsymbol{f}}$ such that

$$1 = \bar{\boldsymbol{f}}^{\mathrm{T}} M\bar{\boldsymbol{f}}\,. \tag{II.46}$$

This can be seen directly form the normalisation condition in Eq. (II.30) combined with the following calculation:

$$1 = \int_{-\infty}^{\infty} \mathrm{d}t\, f^2(t) \approx \sum_{i=1}^{N_{\mathrm{bs}}-2}\sum_{j=1}^{N_{\mathrm{bs}}-2} f_i f_j B(u_i^{(d)}, u_j^{(d)}) = \sum_{i=1}^{N_{\mathrm{bs}}-2}\sum_{j=1}^{N_{\mathrm{bs}}-2} f_i M_{i,j} f_j = \bar{\boldsymbol{f}}^{\mathrm{T}} M\bar{\boldsymbol{f}}\,. \tag{II.47}$$

Finally, to solve the generalised matrix eigenvalue problem posed by Eq. (II.45) `AlkCalc` employs the routine DSBGVX from the software collection LAPACK [10] taken from *Netlib* [9]. This routine is specifically designed to solve real-symmetric banded generalised eigenvalue problems. It further allows to extract eigenvalues selectively. This is useful when the dimension $N_{\mathrm{bs}} - 2$ of the eigenvalue problem is much larger than the number of eigenstates that shall be computed.

At this point there are two open questions to be answered. **(i)** The generalised matrix eigenvalue problem posed by Eq. (II.45) is an approximation to the exact eigenvalue problem posed by Eq. (II.27). One might ask if the numerically computed eigenvalues and eigenvectors indeed approximate the exact result. For instance, on a glimpse one might think that it is possible to find eigenvalues which are not physical but sole numerical artefacts of the approximate expansion of f in a finite set of basis states. **(ii)** The second question is how the components of the matrices K, M, and W can be calculated numerically in an efficient manner, as these are given by integrals that, generally, need to be solved numerically.

(i) The first question is readily answered by the Rayleigh-Ritz variational method. For a detailed discussion on this method please see Ref. [6]. In short, the solutions to the eigenvalue problem posed by Eq. (II.24) are given by the (normalised) stationary points of the following functional:

$$\mathcal{R}[R] = \frac{\langle R, H_{l,s,j} R\rangle}{\langle R, R\rangle}\,. \tag{II.48}$$

The Rayleigh-Ritz variational method is useful in our case because it can be shown that in the finite B-spline basis employed by `AlkCalc` to approximate the radial eigenfunctions f, the

variational principle results in upper bounds for the eigenenergies and good approximations of the radial eigenfunctions [6]. This is useful because, in the finite B-spline basis, solving $\delta\mathcal{R}[R_0] = 0$ for the stationary points R_0 of the functional (II.48) is equivalent to the generalised matrix eigenvalue problem posed by Eq. (II.45). In fact, it be shown that each solution of the generalised matrix eigenvalue problem converges to the exact eigenvalue and the exact radial eigenstate as the number of knots (or, analogously N) is increased. Therefore, one can be sure that the solutions found by solving the generalised matrix eigenvalue problem indeed approximate the exact radial eigenenergies $E_{n,l,s,j}$ and eigenstates $R_{n,l,s,j}$, without producing solutions that do not converge to solutions of the exact eigenvalue problem for the radial eigenstates $R_{n,l,s,j}$.

(ii) To numerically solve the integrals that define the components of the matrices K, M, and V `AlkCalc` employs so-called *Gauß-Legendre quadratures* [11]. This method works as follows: the integral of a (sufficiently well-behaved) function y over the interval $[a, b]$ ($a \leq b$) can be approximated as follows:

$$\int_a^b \mathrm{d}x\, y(x) \approx \sum_{i=1}^{n} \frac{b-a}{2} w_i y\left(\frac{b-a}{2} x_i + \frac{b+a}{2}\right). \tag{II.49}$$

Here n is the so-called *order* of the Gauß-Legendre quadrature, w_i are weights, and x_i points in the open interval (a, b). Importantly, for a quadrature of order n there exist weights w_i and points x_i such that the for all polynomials of degree $d \leq 2n-1$ Eq. (II.49) is exact, i.e., the approximation sign is an equal sign. To obtain the weights w_i and the points x_i the software `AlkCalc` employs the software package GAUSSQ [12] from *Netlib* [9]. Using such a quadrature rule, the components of the matrices K, M, and W can be computed efficiently. To compute the components of the stiffness matrix K one chooses the quadrature order $n = d$, because the derivative of a B-spline is a piecewise polynomial of degree $d-1$. Of course, computing the components of K is only sensible for $d \geq 2$, because otherwise the second derivative vanishes. In practice, this is not a restriction because typical B-spline orders ($k = d+1$) are at least k~6 or higher. To compute the components of the matrix M one chooses the quadrature order $n = d + 1$. With this choice polynomials of degree $2(d+1) - 1 = 2d + 1$ are integrated exactly, which is enough, considering that the relevant integrals are over polynomials of degree $2d$. Finally, the components of the matrix W representing the effective single-electron potential must be calculated. The polynomial order of the integrand is $2d + d_v$, where d_v is the polynomial order of the effective potential (II.37). Of course, in general, this potential is not a polynomial. However, when the knots t_i are placed sufficiently close, on each interval $[t_i, t_{i+1})$ the potential v is approximately a polynomial. This results from the fact that v is an analytical function on the open interval $(0, t_{\max})$. However, to obtain a good approximation it is best to choose a high quadrature order. The employed quadrature order is hard-coded to retain simplicity, but of course can be directly viewed or changed in the source code of `AlkCalc` .

iii. Radial matrix elements

To compute radial matrix elements one calculates for $p \in \mathbb{R}$ (such that the integral exists)

$$\frac{r_p^{n,l,s,j;n',l',s',j'}}{a_\mathrm{B}^p} \equiv \frac{1}{a_\mathrm{B}^p}\langle R_{n,l,s,j}, \mathrm{id}^p R_{n',l',s',j'}\rangle = \int_0^\infty \mathrm{d}r\, r^2 R_{n,l,s,j}(r) \frac{r^p}{a_\mathrm{B}^p} R_{n',l',s',j'}(r)$$

$$= \int_0^\infty (\mathrm{d}t\, a_\mathrm{B})(a_\mathrm{B}t)^2 f_{n,l,s,j}(t) \frac{(a_\mathrm{B}t)^p}{a_\mathrm{B}^p} \frac{1}{a_\mathrm{B}^3} \frac{f_{n',l',s',j'}(t)}{t^2}$$
$$\approx \int_I (\mathrm{d}t\, a_\mathrm{B})(a_\mathrm{B}t)^2 f_{n,l,s,j}(t) \frac{(a_\mathrm{B}t)^p}{a_\mathrm{B}^p} \frac{1}{a_\mathrm{B}^3} \frac{f_{n',l',s',j'}(t)}{t^2} = B(f_{n,l,s,j}, \mathrm{id}^p f_{n',l',s',j'})\,. \tag{II.50}$$

Here, 'id' denotes the identity map. The remaining integral can be approximated with solutions of the generalised matrix eigenproblem posed by Eq. (II.45), i.e., $f_{n,l,s,j}$ and $f_{n',l',s',j'}$ are represented by the vectors $\boldsymbol{f}_{n,l,s,j}$ and $\boldsymbol{f}_{n',l',s',j'}$, respectively [see Eq. (II.38)]. Within this approximation

$$B(f_{n,l,s,j}, \mathrm{id}^p f_{n',l',s',j'}) = \bar{\boldsymbol{f}}^\mathrm{T}_{n,l,s,j} R_p \bar{\boldsymbol{f}}_{n',l',s',j'}\,, \tag{II.51}$$

where $\bar{\boldsymbol{f}}_{n,l,s,j}$ denotes the coefficient vector $\boldsymbol{f}_{n,l,s,j}$ stripped of its first and final component. Further, the components of the real symmetric matrix R_p are defined as

$$(R_p)_{k,k'} = B(\phi_k, \mathrm{id}^p \phi_{k'}) = \begin{cases} B(\phi_k, \mathrm{id}^p \phi_{k+1}), & k' = k+1 \text{ or } k = k'+1 \ (0 < k, k' < N-2) \\ B(\phi_k, \mathrm{id}^p \phi_k), & k' = k \ (0 < k, k' \leq N-2) \\ 0, & \text{else} \end{cases}\,. \tag{II.52}$$

To numerically compute the these components the software `AlkCalc` again employs Gauß-Legendre quadratures defined in Eq. (II.49). For integer $p \geq 0$ the relevant integrals are over polynomials of order $2d + p$. Therefore, one employs the quadrature of order $n = d + 1 + p/2$ for even p and $n = d + (p+1)/2$ for odd ones. When p is not an integer, one chooses sufficiently large n such that the integral is well approximated. The indeed employed quadrature order is hard coded to retain simplicity, but of course can be directly viewed or changed in the source code of `AlkCalc`.

iv. Oscillator strengths

The so-called *oscillator strength* of a transition from an initial state $\boldsymbol{\Psi}_\mathrm{i} \equiv \boldsymbol{\Psi}_{n,l,s,j,m_j}$ to a final state $\boldsymbol{\Psi}_\mathrm{f} \equiv \boldsymbol{\Psi}_{n',l',s',j,m'_j}$ is [6]

$$f_{\mathrm{i}\to\mathrm{f}} = \sum_{m'_j=-j'}^{j'} \frac{2}{3} \bar{E}_{\mathrm{f,i}} |\langle \boldsymbol{\Psi}_\mathrm{f}, \boldsymbol{r}/a_\mathrm{B} \boldsymbol{\Psi}_\mathrm{i} \rangle|^2\,, \tag{II.53}$$

with the energy difference $E_\mathrm{H}\bar{E}_{\mathrm{f,i}} = E_\mathrm{f} - E_\mathrm{i}$. Note that the sum runs over all final m'_j because one is interested in transitions between the fine-structure states without selecting Zeeman sublevels. The oscillator strength is a measure for how strongly the induced Hertzian (electric) dipole oscillates: the radiation power is large for spatially large dipoles and fast oscillation frequencies, captured by the matrix element and energy difference, respectively. In principle, all the machinery to numerically compute this quantity is already there, however, one can reduce the expression to a much simpler form by using angular momentum algebra, in particular, relations from [13]. To simplify the expression first note (Condon-Shortley phase convention)

$$r_x = \sqrt{\frac{2\pi}{3}} r (Y_{1,-1} - Y_{1,1})\,,\ r_y = \mathrm{i}\sqrt{\frac{2\pi}{3}} r (Y_{1,-1} + Y_{1,1})\,,\ r_z = \sqrt{\frac{4\pi}{3}} r Y_{1,0}\,. \tag{II.54}$$

(l, j)	$\mapsto$	(l', j')	a_l
$(l, l+s)$	$\mapsto$	$(l+1, l+s)$	$\frac{1}{(2l+3)(2l+1)}$
$(l, l-s)$	$\mapsto$	$(l+1, l+s)$	$\frac{l+1}{2l+1}$
$(l, l+s)$	$\mapsto$	$(l+1, l+3s)$	$\frac{l+2}{2l+3}$
$(l, l-s)$	$\mapsto$	$(l-1, l-s)$	$\frac{1}{(2l+1)(2l-1)}$
$(l, l+s)$	$\mapsto$	$(l-1, l-s)$	$\frac{l}{2l+1}$
$(l, l-s)$	$\mapsto$	$(l-1, l-3s)$	$\frac{l-1}{2l-1}$

Table 1: Angular factors. Table containing the angular factors a_l [see Eq. (II.57)] for computing oscillator strengths. Throughout this table, $s = 1/2$.

With this, the oscillator strength becomes

$$f_{\mathrm{i}\to\mathrm{f}} = \frac{4\pi}{3}\frac{2}{3}\bar{E}_{\mathrm{f,i}}\langle rR_{\mathrm{f}}, r^2/a_{\mathrm{B}}rR_{\mathrm{i}}\rangle^2 \sum_{m'_j=-j'}^{j'}\sum_{q=-1}^{1}\left|\langle \mathbf{\Phi}_{\mathrm{f}}, Y_{1,q}\mathbf{\Phi}_{\mathrm{i}}\rangle\right|^2 . \tag{II.55}$$

Employing relations from [13, p. 236, 244, 260, 445, 496] one can show that the remaining matrix element is

$$\langle \mathbf{\Phi}_{\mathrm{f}}, Y_{1,q}\mathbf{\Phi}_{\mathrm{i}}\rangle = (-1)^{j'+j+1+l'+s-m'_j}\sqrt{\frac{3}{4\pi}}\begin{pmatrix} j' & 1 & j \\ -m'_j & q & m_j\end{pmatrix}\sqrt{(2j+1)(2j'+1)(2l+1)}\begin{Bmatrix} l & s & j \\ j' & 1 & l'\end{Bmatrix}C^{l,0,1,0}_{l',0}, \tag{II.56}$$

expressed in terms of a Wigner $3jm$-symbol, a Wigner $6j$-symbol, and a Clebsch-Gordan coefficient. With symmetry relations and a rule for summation from [13, p. 236, 259] one can compute the sums over q and m'_j in the expression for the oscillator strength. This yields

$$\sum_{m'_j} f_{\mathrm{i}\to\mathrm{f}} = \frac{2}{3}\bar{E}_{\mathrm{f,i}}\left|\langle rR_{n',l',s,j'}, r/a_{\mathrm{B}}R_{n,l,s,j}\rangle\right|^2 a_l\,,\ a_l = l_{\max}(2j'+1)\begin{Bmatrix} l & s & j \\ j' & 1 & l'\end{Bmatrix}^2, \tag{II.57}$$

where $l_{\max} = \max(l, l')$. There are only six transitions for which a_l is non-zero. Using relations from [13, p. 298, 300] one arrives at the results shown in Tab. 1. With this it is now straight forward to compute the oscillator strengths numerically with a computer program. As a final remark, note that the oscillator strength does not depend on m_j, i.e., after summing over final states, the oscillators for different m_j all have the same strength.

v. State lifetimes

The lifetime τ of a fine-structure state is given by the formula

$$\tau = \Gamma^{-1}\,, \tag{II.58}$$

where Γ is the total rate at which the state is depopulated. At sufficiently low temperatures this rate is given by the sum over the so-called *Einstein-A* coefficients $A_{\mathrm{i}\to\mathrm{f}}$ [6, 14]. These coefficients

are the rates at which the state of interest, here referred to as the 'initial state', radiatively decays to some final state. In the dipole approximation the sum over Einstein-A coefficients can be expressed as [6]:

$$A_\mathrm{i} = \sum_\mathrm{f} A_{\mathrm{i}\to\mathrm{f}}\,,\; A_{\mathrm{i}\to\mathrm{f}} = \frac{E_\mathrm{H}}{\hbar}\frac{4}{3}\alpha^2 \bar{E}_{\mathrm{i,f}}^3 \sum_{u=x,y,z} \left|\langle \mathbf{\Psi}_\mathrm{f}, r_u/a_\mathrm{B}\mathbf{\Psi}_\mathrm{i}\rangle\right|^2 . \tag{II.59}$$

In terms of oscillator strengths one obtains

$$A_\mathrm{i} = \frac{2\alpha^3 E_\mathrm{H}}{\hbar} \sum_{n',l',j'} \bar{E}_{\mathrm{i,f}}^2 (-f_{\mathrm{i}\to\mathrm{f}})\,. \tag{II.60}$$

The rate A_i in Eq. (II.60) is the spontaneous emission rate, which is only non-zero for transitions where a photon is emitted and the final state's energy is lower than the energy of the initial state, i.e., $E_\mathrm{i} > E_\mathrm{f}$. Note that this restricts the quantum numbers in the sum.

At sufficiently high temperatures the spectral density of the environment cannot be neglected. To approximate the spectral density of the environment one can employ Plank's famous formula [15] for the spectral density of a black-body. This yields an average number of photons at frequency ω and temperature T:

$$n(\omega, T) = \frac{1}{\exp(\hbar\omega/(k_\mathrm{B}T)) - 1}\,. \tag{II.61}$$

Here, k_B is Boltzmann's constant. Compared to the low temperature regime, the additional photons at temperature T stimulate the emission of photons and thereby the decay of the initial state to lower-energy final states. Even depopulation of the initial state is possible by the absorption of photons from the environment. This process drives transitions to final states which are higher in energy than the initial state. According to Einstein's rate equations [14] the total rate for depopulation of the initial state is

$$\frac{\hbar}{E_\mathrm{H}}\Gamma = 2\alpha^3 \sum_{n',l',j'} \bar{E}_{\mathrm{i,f}}^2 (-f_{\mathrm{i}\to\mathrm{f}})(1 + n(E_{\mathrm{i,f}}/\hbar, T)) + 2\alpha^3 \sum_{n',l',j'} \bar{E}_{\mathrm{f,i}}^2 f_{\mathrm{i}\to\mathrm{f}}\, n(E_{\mathrm{f,i}}/\hbar, T)\,, \tag{II.62}$$

where each sum is restricted to quantum numbers for the summands are positive. This entails that the first sum runs only over final states with lower energy compared to the initial state; this describes the emission of photons. The second sum runs solely over final states with larger energy than the initial state; this corresponds to the case of photon absorption.

III. Manual

Reference manual for the library functions of `AlkCalc` . Prototypes for the following functions are defined in the header `interface/alkcalc.h`.

There is always a decision to be made when dealing with half-integer quantum numbers numerically. Here, the following agreement is made: say q is a quantum number that may take half-integer values, e.g., j, m_j. To 'clean up' the numerical value it is understood that $2q$ is equivalent to $[2.0 \times q + 0.5]$. The 'floor' function is defined as $[x] = k \in \mathbb{Z}$ ($x \in \mathbb{R}$), where k is the unique integer such that $x \in [k, k+1)$. In practice, this means that for any $q \in [-0.75, -0.25)$, $q = -1/2$ is used, for any $q \in [-0.25, 0.25)$, $q = 0$, for any $q \in [0.25, 0.75)$, $q = 1/2$, and so on, in both the negative and positive directions. In the following manual, this convention is applied to all quantum numbers that can take half-integer values, unless stated otherwise.

FUNCTION

`alkcalc_Enlsj`

DESCRIPTION

Eigenenergies in units of Hartree. The requested eigenenergy is read from the corresponding data file in the directory `data` and returned.

ARGUMENTS

`char *species`
String to specify atom/ion species, e.g., 1H for Hydrogen, or 88SR+ for the $^{88}Sr^{+}$ ion.

`int32_t n`
Principal quantum number.

`int32_t l`
Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$.

`double s`
Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double j`
Total angular momentum quantum number $j = |l - 1/2|, l + 1/2$.

RETURN

`double E`
E is the eigenenergy specified by the quantum numbers n, l, $s = 1/2$, and j.

FUNCTION

`alkcalc_fnlsj`

DESCRIPTION

Numerical representation $\boldsymbol{f}_{n,l,s,j}$ of the radial eigenstate $R_{n,l,s,j}$ from Eq. (II.25). $R_{n,l,s,j}$ is recovered via Eq. (II.38) and the parametrisation (II.26). The result is owned by the caller and should be freed with `alkcalc_state_free`.

ARGUMENTS

`char result`

Character for deciding if knotdata is returned; see RETURN down below. If `result` is `f` (full), the full result is returned, and if `result` is `p` (partial), the knotdata is not returned.

`char *species`

String to specify atom/ion species, e.g., `1H` for Hydrogen, or `88SR+` for the $^{88}Sr^{+}$ ion.

`int32_t n`

Principal quantum number.

`int32_t l`

Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$.

`double s`

Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double j`

Total angular momentum quantum number $j = |l - 1/2|, l + 1/2$.

RETURN

`alkcalc_state *f`

`f` contains ten members: `n`, `l`, `j`, `k`, `Nks`, `N`, `Nbs`, `t`, `h`, `fnlsj`.

`int32_t n`: Principal quantum number, n.
`int32_t l`: Orbital angular momentum quantum number, l.
`double j`: Total angular momentum quantum number, j.
`int32_t k`: Order of B-splines, $k = d + 1$ (d polynomial degree of B-splines).
`int32_t Nks`: Total number of knots, N_{ks}.
`int32_t N`: Number of knots (no multiplicities), N.
`int32_t Nbs`: Number of B-splines, N_{bs}.
`double *t`: Array of length N_{ks} containing the knots $(t_i)_{i=0,\ldots,N_{\mathrm{ks}}-1}$.
`double *h`: Array of length $N-1$ containing $h_i = t_{i+k} - t_{i+k-1}$, $i = 0, \ldots, N-2$.
`double *fnlsj`: Array of length N_{bs} containing the components of $\boldsymbol{f}_{n,l,s,j}$.

FUNCTION

`alkcalc_fnlsj_eval`

DESCRIPTION

Function to evaluate the radial eigenfunction $f_{n,l,s,j}$ defined in Eq. (II.38) at the points specified in the array `tevals`. Note that via the parametrisation (II.26) this function directly allows to evaluate the radial eigenstates $R_{n,l,s,j}(r)$ at any r. The result is stored in the array `tevals`, i.e., the function `alkcalc_fnlsj_eval` overwrites each element t in `tevals` with $f_{n,l,s,j}(t)$.

ARGUMENTS

`char *species`
String to specify atom/ion species, e.g., `1H` for Hydrogen, or `88SR+` for the $^{88}Sr^{+}$ ion.

`int32_t n`
Principal quantum number.

`int32_t l`
Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$.

`double s`
Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double j`
Total angular momentum quantum number $j = |l - 1/2|, l + 1/2$.

`double *tevals`
Array containing the points $t \in [0, t_{\max}]$ at which to evaluate the function $f_{n,l,s,j}$. After the function call, the entries t of the array `tevals` are overwritten with the associated function values $f_{n,l,s,j}(t)$.

`int32_t ltevals`
Length of the array `tevals`.

FUNCTION

`alkcalc_rp`

DESCRIPTION

Radial matrix element of r^p, $p \in \mathbb{R}$ [see Eq. (II.50)]. It is the user's responsibility to ensure that the integral associated to the desired matrix element exists. In case it does not, the returned value is wrong and no error is thrown. ^{1}H (Hydrogen atom) and ^{2}He$^+$ (Helium ion) can be used to benchmark atoms and ions, respectively.

ARGUMENTS

`char *species`
String to specify atom/ion species, e.g., 1H for Hydrogen, or 88SR+ for the ^{88}Sr$^+$ ion.

`int32_t nb`
Principal quantum number of bra.

`int32_t lb`
Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$ of bra.

`double sb`
Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double jb`
Total angular momentum quantum number of bra, j.

`double p`
Power of radius operator; cf. Eq. (II.50). For instance, $p = 0$ yields the scalar product of the states. For $p = 1$ and $p = 2$, respectively, dipole and quadrupole matrix elements are computed. Generally, $p \in \mathbb{R}$.

`int32_t nk`
Principal quantum number of ket.

`int32_t lk`
Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$ of ket.

`double sk`
Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double jk`
Total angular momentum quantum number of ket, j.

RETURN

`double rp`
$r_p^{n,l,s,j;n',l',s',j'}$ in Eq. (II.50) in units of a_B^p, where a_B is Bohr's radius. Quantum numbers with a prime correspond to the bra, and the ones without one, to the ket.

FUNCTION

`alkcalc_cj1m1j2m2jmj`

DESCRIPTION

Clebsch-Gordan coefficients, $C^{j_1,m_1,j_2,m_2}_{j,m_j} = \langle j_1, m_1; j_2, m_2 | j_1, j_2, j, m_j \rangle$, where j is the total angular momentum composed of the two individual ones j_1 and j_2. To fix the phase uniquely, the Condon-Shortley sign convention is adopted, that is, the Clebsch-Gordan coefficients with $m_j = j_1 + j_2$ are positive. In particular, for the case $j_1 = l$ and $j_2 = s$,

$$C^{l,m_l,s,m_s}_{j,m_j} = \langle Y_{l,m_l} \otimes \boldsymbol{\chi}_{s,m_s}, \boldsymbol{\Phi}_{l,s,j,m_j} \rangle \,, \tag{III.1}$$

with the states $Y_{l,m_l} \otimes \boldsymbol{\chi}_{s,m_s}$ of the uncoupled basis and the coupled basis states $\boldsymbol{\Phi}_{l,s,j,m}$; see Eqs. (II.12, II.16), and Eq. (II.17), respectively. The return type (see RETURN down below) allows to extract the requested coefficient symbolically as well as numerically. The function returns zero in case the arguments do not make sense, e.g., if $j_1 < m_1$; or if j_1 is half-integer and m_1 is integer; or if j does not lay between (including bounds) $|j_1 - j_2|$ and $|j_1 + j_2|$.

ARGUMENTS

`double j1`
Angular momentum quantum number of first factor.

`double m1`
Magnetic quantum number of first factor.

`double j2`
Angular momentum quantum number of second factor.

`double m2`
Magnetic quantum number of second factor.

`double j`
Total angular momentum quantum number.

`double mj`
Magnetic quantum number of total angular momentum.

RETURN

`alkcalc_cg c`
`c` contains three members: `sign`, `numerator`, and `denominator`.
`int8_t sign`: Sign, σ, of the Clebsch-Gordan coefficient.
`int64_t numerator`: Numerator, ν, of the Clebsch-Gordan coefficient.
`int64_t denominator`: Denominator, μ, of the Clebsch-Gordan coefficient.
The requested Clebsch-Gordan coefficient is given by $\sigma\sqrt{\nu/\mu}$. Euklid's algrithm is employed to reduce ν and μ. This representation allows to obtain Clebsch-Gordan coefficients symbolically as well as numerically.

FUNCTION

`alkcalc_YlmlXsms`

DESCRIPTION

Angular factor of the full state in the uncoupled basis, i.e., the two-component spinor $Y_{l,m_l}(\vartheta,\varphi)\boldsymbol{\chi}_{s,ms}$. The input quantum numbers are checked for inconsistencies and an error is thrown in case one is detected. The angles ϑ and φ have the ranges $[0,\pi]$ and $[0,2\pi]$, respectively. This condition is checked up to floating point machine precision (type `double`) and an error will be thrown in case either condition is not obeyed.

ARGUMENTS

`int32_t l`
Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$.

`int32_t ml`
Magnetic quantum number of orbital angular momentum.

`double s`
Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double ms`
Magnetic quantum number of electron spin, m_s. For any $m_s < 0$ the value $m_s = -1/2$ is used internally, and for $m_s \geq 0$ the value $m_s = 1/2$.

`double theta`
Polar angle (Zenitwinkel), ϑ, in range $[0,\pi]$.

`double phi`
Azimuthal angle (Azimut), φ, in range $[0,2\pi]$.

RETURN

`alkcalc_spinor s`
`s` has two components: `u` and `v`.
`u` is the spin-up ($m_s = 1/2$) component.
`d` is the spin-down ($m_s = -1/2$) component.

FUNCTION

`alkcalc_Philsjmj`

DESCRIPTION

Angular factor of the full eigenstate in the coupled (fine-structure) basis, i.e., the two-component spinor $\Phi_{l,s,j,mj}(\vartheta, \varphi)$ [see Eq. (II.17)]. The input quantum numbers are checked for inconsistencies and an error is thrown in case one is detected. The angles ϑ and φ have the ranges $[0, \pi]$ and $[0, 2\pi]$, respectively. This condition is checked up to floating point machine precision (type `double`) and an error will be thrown in case either condition is not obeyed. To compute the resulting two-component spinor, Clebsch-Gordon coefficients are needed, for which the sign convention detailed in Eq. (III.1) is employed.

ARGUMENTS

`int32_t l`
Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$.

`double s`
Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double j`
Total angular momentum quantum number $j = |l - 1/2|, l + 1/2$.

`double mj`
Magnetic quantum number of total angular momentum, m_j.

`double theta`
Polar angle (Zenitwinkel), ϑ, in range $[0, \pi]$.

`double phi`
Azimuthal angle (Azimut), φ, in range $[0, 2\pi]$.

RETURN

`alkcalc_spinor s`
`s` has two components: `u` and `v`.
`u` is the spin-up ($m_s = 1/2$) component.
`d` is the spin-down ($m_s = -1/2$) component.

FUNCTION

`alkcalc_fitof`

DESCRIPTION

Oscillator strength for the transition between an initial state Ψ_i and a final state Ψ_f, as defined in Subsec. iv, that is, it was already summed over the magnetic quantum number of the final state.

ARGUMENTS

`char *species`
String to specify atom/ion species, e.g., 1H for Hydrogen, or 88SR+ for the $^{88}Sr^{+}$ ion.

`int32_t ni`
Principal quantum number of the initial state Ψ_i.

`int32_t li`
Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$ of the initial state Ψ_i.

`double si`
Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double ji`
Total angular momentum quantum number of the initial state Ψ_i, j.

`int32_t nf`
Principal quantum number of the final state Ψ_f.

`int32_t lf`
Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$ of the final state Ψ_f.

`double sf`
Electron spin quantum number of the final state Ψ_f. Not an argument here, since $s = 1/2$ is always assumed.

`double jf`
Total angular momentum quantum number of the final state Ψ_f, j.

RETURN

`double fitof`
$f_{\mathrm{i}\to\mathrm{f}}$ is the oscillator strength of the selected electronic transition, summed over m'_j of the final state. Note that because of this sum, it is the same for all m_j of the initial state.

FUNCTION

`alkcalc_tau`

DESCRIPTION

Lifetime of the fine-structure state Ψ_{n,l,s,j,m_j} in nanoseconds. Note that, since the oscillator-strengths [see Eq. (II.57)] do not depend on m_j, also the lifetimes do not depend on it [see Eq. (II.62)].

ARGUMENTS

`double T`

Temperature of black-body excitation spectrum in Kelvin.

`char *species`

String to specify atom/ion species, e.g., 1H for Hydrogen, or 88SR+ for the $^{88}Sr^+$ ion.

`int32_t n`

Principal quantum number.

`int32_t dn`

Range for principal quantum number. At zero temperature, this argument is not used. For finite temperature, the absorption processes in Eq. (II.62) allow 'decay' to states with higher energy. In principle, one needs to take into account all transitions to these states, of which there are infinitely many. Here, solely `dn` levels above Ψ_{n,l,s,j,m_j} are taken into account, in order to truncate the series, for both $l' = l \pm 1$. In other words, the sum in Eq. (II.62), taking into account absorption, contains $2 \times$ `dn` terms for $l > 0$ and `dn` for $l = 0$.

`int32_t l`

Orbital angular momentum quantum number $l = 0, 1, \ldots, n-1$.

`double s`

Electron spin quantum number. Not an argument, as $s = 1/2$ is always assumed.

`double j`

Total angular momentum quantum number $j = |l - 1/2|, l + 1/2$.

RETURN

`double tau`

τ is the lifetime of the fine-structure state Ψ_{n,l,s,j,m_j} (independent of m_j) in nanoseconds.

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